Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/16/2006
PI Name:	DeLuca, Dominick Ph.D.		
Project Title:	The Effects of Vector-Averaged Gravity on the Development of T cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	deluca@u.arizona.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	520-621-8941
Organization Name:	University of Arizona		
PI Address 1:	LSN Room 648		
PI Address 2:	1501 N Campbell Avenue		
PI Web Page:			
City:	Tucson	**State:**	AZ
Zip Code:	85724	**Congressional District:**	7
Comments:	PI is deceased as of 10/26/2007 (info from A. Chu/ARC; 9/16/2008)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	04/01/2004	**End Date:**	10/26/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/26/2007 when Dr. DeLuca passed away. (per A. Chu/ARC 9/2008) NOTE: Received NCE to 9/30/2008 per A. Chu/ARC (6/07) NCE to 9/30/2007, per A. Chu 6/23/06		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC97G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We have developed a microgravity organ culture dish system (MOCDS) that allows for the normal development of T cells from hematopoietic stem cell (HSC) progenitors. This system mimics the normal development of T cells outside of the body, away from stress hormones made during flight. The MOCDS can be used both on the clinostat and during spaceflight. The culture systems are easy to set up and manipulate which allows for a more detailed analysis. Using this specialized system, our work has revolved around studying the effects of vector-averaged gravity on the development of T cells. In our previous work that utilized MOCDS we demonstrated the gravity-dependence of T cell development. The results from this work suggest that the full sequelae of pre-TCR signaling may not be taking place in vector-averaged gravity, as indicated by a block at the immature single positive stage of development. Since then, we have set out to determine what the underlying mechanism was for this block. Our current focus has been on analyzing the cytokine profiles for our clinorotated cultures, specifically TNF-alpha and investiagtion of the exact point of blockage using pT-alpha and proliferation studies. Using the information obtained from these and our other studies, we aim to better understand, and hopefully reverse, the effects of clinorotation. Task Objective: Determine the mechanism by which vector-averaged gravity blocks T cell development at the ISP developmental stage using organ culture, which preserves the 3-D microenvironment of the tissue. Task Significance: This will allow for better understanding of the mechanism underlying decreased T cell development in vector-averaged gravity. This knowledge can be applied to reversing the effects of vector-averaged gravity on developing T cells. A functional immune system is critical if humans intend to reside in environments that have minimal gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work has shed light on the signal dampening effects of microgravity exposure. By further understanding the underlying mechanism of this microgravity-induced effect, we could initiate a new line of research centered on the treatment and prevention of other thymocyte developmental problems, such as Type I diabetes. Many T cell developmental problems involve signaling defects similar to those seen in our microgravity work. For instance, microgravity exposure led to a significant increase in the production of TNF-alpha. Other work investigating the role of TNF-alpha during T cell development has shown that over expression of this cytokine may lead to deficient negative selection and hence the production of potentially autoreactive T cells.
Task Progress:	The article describing the majority of our was published in Developmental and Comparative Immunology. Generally, this work shows that after 4 days of vector-averaged gravity exposure, the fetal thymus organ cultures (FTOC) exhibited a significant reduction in CD4+CD8+ double positive (DP) cell production. Furthermore, CD4-CD8- double negative (DN) cells expressed lower levels of CD3 on their surface. DN, as well as immature single positive (ISP) cells, also expressed reduced levels of the IL-7 receptor a chain (CD127). These reductions in CD3 and CD127 expression were concomitantly associated with an increased production of tumor necrosis factor (TNF)-alpha (a). We were also able to show that addition of an exogenous signal (anti-CD3e monoclonal antibody) to these cultures effectively mitigated the MMG-induced effects, therefore suggesting that MMG-exposure, does in fact, cause a signal dampening effect on developing thymocytes. Last year we further analyzed the increased production of tumor necrosis factor (TNF)-a in MMG. Its increased presence in clinorotated cultures suggests that it plays an important role in the development of DP thymocytes. To analyze further, we added soluble TNF receptor (1ug) to three day cultures and found that double positive cell production was increased in both the clinostat and motional control. More specifically, the addition of the receptor to the clinorotated culture was able to increase the level of DP thymocytes and decrease the level of DN thymocytes to a level comparable to the motional control without TNF receptor. Interestingly, the motional control showed a higher DN population compared to the DP thymocyte population, but with the addition of TNF receptor, these values changed to levels similar to the stationary control. This suggests that the vibrations of the motional control are aiding to the decreased DP development seen in the clinorotated cultures. The stepwise progression of increased DP thymocytes resulting from the addition of TNF receptor in both the clinostat and the motional control, coincided with a decrease in CD44+ CD25+ population. This DN subset of cells was found to make up the majority of DN cells in each of the dishes. To further analyze the effects of TNFa, we have sought to establish the primary producers of TNFa and to pinpoint the exact stage in lymphocyte development that is being effected by microgravity. Internal staining analysis performed during this year has indicated that the production of TNFa is elevated, not only in the developing lymphocyte populations, but also in the thymic epithelial and antigen presenting cells that create the lymphocyte selection environment. In fact, TNFa production is more prevalent and more elevated in non-lymphoid tissues vs. lymphoid tissues, suggesting a disruption of the normal function of the thymic milieu. The fact that TNFa is a known TCR signal-dampening agent, suggests TNFa may be significantly reducing signals leading to stagnation during development. Thus, we are currently pinpointing the stagnation through the use of time dependent pTa staining and Brd-U incorporation experiments, so that further TCR and TNFa signaling pathway analysis can be performed at the peak of inhibition.
Bibliography Type:	Description: (Last Updated: 07/05/2019)
Articles in Peer-reviewed Journals	Woods CC, Banks KE, Lebsack TW, White TC, Anderson GA, Maccallum T, Gruener R, DeLuca D. "Use of a microgravity organ culture dish system to demonstrate the signal dampening effects of modeled microgravity during T cell development." Dev Comp Immunol. 2005;29(6):565-82. Epub 2005 Jan 24. PMID: 15752552 , Jun-2005